NAME: PRADIP KALE
ROLL_NO:13224
BATCH: B1
Contents for Theory:
7. Text Analytics
1. Extract Sample document and apply following document preprocessing methods:
Tokenization, POS Tagging, stop words removal, Stemming and Lemmatization.
2. Create representation of document by calculating Term Frequency and Inverse Document
Frequency

In [18]:
```
import nltk
import re
```

In [19]:
```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('averaged_perceptron_tagger')
```

```
[nltk_data] Downloading package punkt to C:\Users\soham
[nltk_data]     kakade\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to C:\Users\soham
[nltk_data]     kakade\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to C:\Users\soham
[nltk_data]     kakade\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to C:\Users\soham
[nltk_data]     kakade\AppData\Roaming\nltk_data...
[nltk_data]   Package omw-1.4 is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     C:\Users\soham kakade\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to[nltk_data]
date!
```

Out[19]: True

In [20]:
```
text = "Tokenization is the first step in text analytics. It breaks text into
```

```
Filtered Words: ['Tokenization', 'first', 'step', 'text', 'analytics', '.',
'breaks', 'text', 'smaller', 'units', '.']
```

In [22]:
```
from nltk.corpus import stopwords

stop_words = set(stopwords.words("english"))

filtered_tokens = []
for word in word_tokens:
    if word.lower() not in stop_words:
        filtered_tokens.append(word)

print("Filtered Words:", filtered_tokens)
```

In [23]:
```
from nltk.stem import PorterStemmer

ps = PorterStemmer()
stemmed = [ps.stem(w) for w in filtered_tokens]

print("Stemmed Words:", stemmed)
```

```
Stemmed Words: ['token', 'first', 'step', 'text', 'analyt', '.', 'break', 'te
xt', 'smaller', 'unit', '.']
```

In [24]:
```
from nltk.stem import WordNetLemmatizer

lemmatizer = WordNetLemmatizer()
lemmatized = [lemmatizer.lemmatize(w) for w in filtered_tokens]

print("Lemmatized Words:", lemmatized)
```

```
Lemmatized Words: ['Tokenization', 'first', 'step', 'text', 'analytics', '.',
'break', 'text', 'smaller', 'unit', '.']
```

In [8]:
```
ps = PorterStemmer()
stemmed = [ps.stem(w) for w in filtered_tokens]
print("\nStemmed Words:", stemmed)
```

```
Stemmed Words: ['text', 'analyt', 'process', 'analyz', 'text', 'data', 'extra
ct', 'use', 'inform', '.']
```

In [25]:
```
pos_tags = nltk.pos_tag(filtered_tokens)
print("POS Tags:", pos_tags)
```

```
POS Tags: [('Tokenization', 'NN'), ('first', 'RB'), ('step', 'VB'), ('text',
'JJ'), ('analytics', 'NNS'), ('.', '.'), ('breaks', 'NNS'), ('text', 'JJ'),
('smaller', 'JJR'), ('units', 'NNS'), ('.', '.')]
```

In [26]:
```
from sklearn.feature_extraction.text import TfidfVectorizer
```

In [27]:
```
doc1 = "Jupiter is the largest planet"
doc2 = "Mars is the fourth planet from the sun"
```

In [28]:
```
documents = [doc1, doc2]
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(docs)
```

In [29]:
```
print("Words:", vectorizer.get_feature_names_out())
```

```
Words: ['analysis' 'analytics' 'analyzing' 'and' 'are' 'data' 'extracts' 'fro
m'
 'important' 'information' 'is' 'mining' 'of' 'process' 'text' 'the'
 'useful']
```

In [30]:
```
import pandas as pd
tfidf_df = pd.DataFrame(X.toarray(), columns=features)
```

In [31]:
```
print("\nTF-IDF Matrix:\n")
print(tfidf_df)
```

```
TF-IDF Matrix:

   analysis  analytics  analyzing       and       are      data  extracts  \
0  0.000000   0.276745   0.363886  0.000000  0.000000  0.276745   0.00000
1  0.000000   0.000000   0.000000  0.000000  0.000000  0.000000   0.39543
2  0.426184   0.324124   0.000000  0.426184  0.426184  0.324124   0.00000

      from  important  information        is   mining        of   process  \
0  0.00000   0.000000       0.00000  0.363886  0.00000  0.363886  0.363886
1  0.39543   0.000000       0.39543  0.000000  0.39543  0.000000  0.000000
2  0.00000   0.426184       0.00000  0.000000  0.00000  0.000000  0.000000

       text       the   useful  0
0.429834  0.363886  0.00000
1  0.467094  0.000000  0.39543
2  0.251711  0.000000  0.00000
```

In [ ]:
```

```